Q). Create structure variable student and estimate size of structure manually. Then write a C programs to find size of student structure using size of program. Now observe both the outputs. Are they same or different? Justify your answer.

A). Structure Definition:

```
struct student
{
  int rollno;
  char grade;
  float marks;
};
```

Size estimation:

int roll = 4 bytes
char grade = 1 byte
float marks = 4 bytes
size = 9 bytes.

C program using size of

```
#include<stdio.h>
 struct student
{
   int rollno;
   char grade;
   float marks;
 };
   int main()
   {
    struct student s;
    printf("size = %zu bytes \n", sizeof(s));
     return 0;
    }
```

o/p : size = 12 bytes

∴ The o/p are different because compiler adds padding b/w structure members so that data is stored at memory addresses that are efficient for processor to access. Although actual data occupies 9 bytes, compiler inserts 3 extra bytes, making it 12 bytes.